

Project Glossary

Key statistical, modeling, and application terms used in the Medical Insurance Cost Prediction project — explained simply, with examples.

Descriptive Statistics

Mean

The average value — add up all numbers and divide by how many there are.

Example: If 5 people pay \$10, \$20, \$20, \$30, \$40 for insurance, the mean is $(10+20+20+30+40)/5 = \$24$.

Median

The middle value when all data points are sorted from smallest to largest.

Example: For charges \$10, \$20, \$20, \$30, \$40, the median is \$20 (the middle one).

Standard Deviation

A measure of how spread out the values are from the mean. A small number means values are close together; a large number means they're spread far apart.

Example: If most people's charges are close to \$13,000, the standard deviation is small. If some pay \$2,000 and others \$60,000, it's large.

IQR (Interquartile Range)

The range covering the middle 50% of the data, found by subtracting the 25th percentile from the 75th percentile. It shows spread while ignoring extreme outliers.

Example: If the 25th percentile of charges is \$4,700 and the 75th percentile is \$16,600, the IQR is about \$11,900.

Skewness

A measure of how lopsided a distribution is. Zero means symmetric; a large positive number means a long tail stretching to the right (high values).

Example: Insurance charges are right-skewed — most people pay a moderate amount, but a few (often smokers) pay very high amounts, stretching the tail to the right.

Kurtosis

A measure of how 'heavy-tailed' or peaked a distribution is compared to a normal (bell-curve) distribution. High kurtosis means more extreme outliers than expected.

Example: High kurtosis in `charges` reflects the unusually high-cost outliers among smokers.

Visual Exploration

Histogram / KDE Plot

A chart showing how often values fall into different ranges (histogram = bars, KDE = a smoothed curve version of the same idea).

Example: A histogram of `age` might show bars for 18-25, 26-35, etc., revealing that ages are fairly evenly spread.

Scatter Plot

A chart plotting two numerical variables against each other, one point per observation, to reveal relationships or patterns.

Example: Plotting `bmi` against `charges` can reveal whether higher BMI tends to come with higher medical costs.

Correlation Matrix

A grid showing how strongly every pair of numerical variables is related, with values from -1 (strong negative) to +1 (strong positive relationship).

Example: A correlation of 0.3 between `age` and `charges` means older age is mildly associated with higher charges.

Hypothesis Testing

Null Hypothesis (H0)

The default assumption that there is NO effect or NO difference between groups. We test the data to see if there's enough evidence to reject this.

Example: H0: 'There is no difference in average charges between smokers and non-smokers.'

Alternative Hypothesis (H1)

The claim that there IS a real effect or difference, which we accept only if the data provides strong enough evidence against H0.

Example: H1: 'Smokers and non-smokers have significantly different average charges.'

Significance Level (alpha, α)

The threshold probability (commonly 0.05) used to decide whether results are 'statistically significant.' If the p-value is below alpha, we reject H0.

Example: At $\alpha = 0.05$, a p-value of 0.03 leads to rejecting H0, but a p-value of 0.20 does not.

P-value

The probability of seeing a result as extreme as what you observed, if H0 were actually true. A small p-value means the observed effect is unlikely to be due to random chance alone.

Example: A p-value of 0.001 for the smoker vs. non-smoker charges test means it's very unlikely this difference happened by chance.

Shapiro-Wilk Test

A statistical test that checks whether a set of numbers is approximately normally distributed (bell-curve shaped). Used to decide which comparison test to run next.

Example: If Shapiro-Wilk gives $p < 0.05$ for `charges` in the smoker group, it means those charges are NOT normally distributed.

Levene's Test

A test that checks whether two or more groups have roughly equal variance (spread). Used to decide the exact settings for a t-test.

Example: Levene's test checks if smokers' charges are just as spread out as non-smokers' charges.

Two-Sample t-test

A test comparing the means of two groups, used when the data in each group is roughly normally distributed.

Example: Used to compare average charges between males and females, if both groups pass the normality check.

Mann-Whitney U Test

A non-parametric alternative to the t-test, used to compare two groups when the data is NOT normally distributed. It compares rankings instead of raw averages.

Example: Used to compare smoker vs. non-smoker charges since charges data is typically skewed, not normal.

Chi-Square Test

A test that checks whether two categorical variables are related/associated, by comparing observed counts to what we'd expect if there were no relationship.

Example: Testing whether `smoker` status is linked to `region` — e.g., are smokers more common in some regions than others?

One-Way ANOVA

A test comparing the means of three or more groups at once, checking if at least one group differs significantly from the others.

Example: Comparing average charges across all four regions (northeast, northwest, southeast, southwest) simultaneously.

Reject / Fail to Reject H_0

The final decision of a hypothesis test. 'Reject H_0 ' means the evidence supports a real difference/effect. 'Fail to Reject H_0 ' means there isn't enough evidence to claim one.

Example: $p = 0.03 < 0.05 \rightarrow \text{Reject } H_0$. $p = 0.40 > 0.05 \rightarrow \text{Fail to Reject } H_0$.

Regression Modeling (OLS)

Linear Regression

A modeling technique that predicts a numeric outcome as a weighted sum of input variables, assuming a straight-line (linear) relationship.

Example: Predicting `charges` using a formula like: $\text{charges} = \beta_0 + \beta_1(\text{age}) + \beta_2(\text{bmi}) + \dots$

OLS (Ordinary Least Squares)

The most common method for fitting a linear regression line — it finds the coefficients that minimize the total squared distance between actual and predicted values.

Example: `statsmodels.api.OLS()` fits the regression line that best predicts `charges` from `age`, `bmi`, `smoker`, etc.

Coefficient (β)

The estimated effect of one predictor on the outcome, holding all other predictors constant.

Example: If the `smoker` coefficient is 23,850, being a smoker is linked to about \$23,850 higher charges, all else equal.

Confidence Interval (95% CI)

A range of plausible values for a coefficient (or prediction). We are 95% confident the true value falls within this range.

Example: A 95% CI of [233, 280] for `age` means we're confident the real effect of age is somewhere in that range.

R-squared (R^2)

The proportion of variation in the outcome variable that the model explains, from 0 (explains nothing) to 1 (explains everything).

Example: $R^2 = 0.75$ means the model explains 75% of why charges vary from person to person.

Adjusted R-squared

A version of R^2 that adjusts for the number of predictors used, penalizing models that add variables without real improvement. More reliable for comparing models.

Example: Adding a random, useless variable to a model can raise R^2 slightly but will lower Adjusted R^2 .

F-statistic

A test statistic that checks whether the regression model as a whole is statistically significant (better than just predicting the average every time).

Example: A very small Prob(F-statistic), like 0.00, confirms the overall model is meaningful, not random noise.

Diagnostic Checks (Gauss-Markov Assumptions)

Residual

The difference between the actual observed value and what the model predicted. $\text{Residual} = \text{Actual} - \text{Predicted}$.

Example: If a person's actual charge is \$15,000 and the model predicted \$13,500, the residual is \$1,500.

Linearity

The assumption that the relationship between predictors and the outcome is a straight line, not curved.

Example: Checked visually — if a Residuals vs. Fitted plot shows a clear curve or pattern, linearity may be violated.

Homoscedasticity

The assumption that residuals have roughly constant spread (variance) across all predicted values, rather than fanning out.

Example: If residuals spread out more for high-cost predictions than low-cost ones, that's a violation (heteroscedasticity).

Residuals vs. Fitted Plot

A diagnostic scatter plot of residuals against the model's predicted values, used to check linearity and homoscedasticity. Random scatter around zero is what you want to see.

Example: A funnel/cone shape in this plot signals that error variance increases as predicted charges increase.

Q-Q Plot (Quantile-Quantile Plot)

A diagnostic plot comparing the distribution of residuals to a theoretical normal distribution. Points falling on the diagonal line indicate normally distributed residuals.

Example: If points curve away from the line at the tails, residuals are not perfectly normal — often due to skewed data.

Jarque-Bera Test

A statistical test that checks whether residuals are normally distributed, based on their skewness and kurtosis.

Example: A low p-value (< 0.05) in the Jarque-Bera test means the residuals significantly deviate from normality.

Multicollinearity

A problem where two or more predictors are highly correlated with each other, making it hard to isolate each one's individual effect.

Example: If `age` and `bmi` moved together almost perfectly, it would be hard to tell which one is really driving charges.

VIF (Variance Inflation Factor)

A number for each predictor showing how much its variance is 'inflated' due to correlation with other predictors. $\text{VIF} > 5$ is often a red flag.

Example: A VIF of 1.5 for `region\_southeast` is low, meaning it isn't strongly overlapping with other predictors.

Application & Deployment

Streamlit

A Python library for quickly building interactive web apps/dashboards without needing separate frontend code (HTML/CSS/JS).

Example: `streamlit run app.py` launches a browser-based dashboard from a single Python script.

Virtual Environment (venv)

An isolated Python environment for a project, keeping its installed packages separate from other projects to avoid version conflicts.

Example: `python -m venv venv` creates an isolated space so this project's package versions don't clash with others on your machine.

requirements.txt

A text file listing every Python package (and often its version) a project needs, so others can install the exact same setup with one command.

Example: Running `pip install -r requirements.txt` installs pandas, streamlit, statsmodels, and everything else listed.

Prediction Interval

A range for a single new individual's predicted outcome, accounting for both model uncertainty and natural randomness. It's wider than a confidence interval.

Example: A 95% prediction interval of [\$8,000, \$22,000] means we're 95% confident THIS specific person's actual charge falls in that range.